AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII EXAMINATION

APRIL/ MAY 2018

Mathematics Paper II

Time: 2 hours 15 minutes Marks: 65

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INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are EIGHT questions. Answer ALL questions. Choices are specified inside the paper.

3. When answering the questions:

   Read each question carefully.
   Use black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a scientific calculator if you wish.

Q.1. (Total 5 Marks)

Determine whether the given function $f(x)$ is continuous at $x = 2$.

$$f(x) = \begin{cases} x+3 & \text{for } -2 \le x \le 2 \\ 2x & \text{for } 2 < x \le 4 \end{cases}$$

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.2.)

Q.2. (Total 12 Marks)

a. Differentiate the following function with respect to x.

i. $y = \text{cosec } 5x$ (2 Marks)

ii. $y = e^{2x} \ln \sqrt{e^x}$ (4 Marks)

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.2.)

b. Expand the Maclaurin series for $\tan x$ up to 3rd term. Show your working.

The Maclaurin expansion for $f(x)$ about $x = 0$ is given as

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \ldots$$ (6 Marks)

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.2.)

c. The volume of a paper box is expressed as $V = 27x - \frac{x^3}{4}$, where the length of a side of the box is x cm.

i. Find the value of x for which the volume of the box is maximum. (2 Marks)

ii. Show that the volume of the box at x is maximum. (3 Marks)

iii. Find the maximum volume of this box. (1 Mark)

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.3.)

Q.3. (Total 12 Marks)

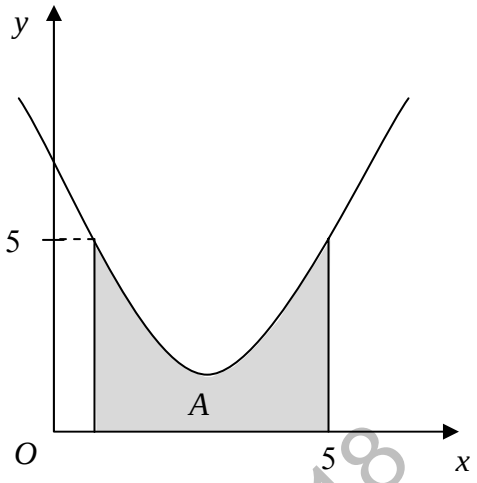
a. Show that $\int \frac{2}{x(x+1)}dx = \ln\left(\frac{x}{x+1}\right)^2 + c$. (6 Marks)

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.3.)

b. The given diagram shows the curve of $y = (x-3)^2 + 1$. Find the area of the shaded region *A*.

(6 Marks)



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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.3.)

c. Evaluate $\int \frac{xdx}{\sqrt{x^2+9}}$. (6 Marks)

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.4.)

Q.4. (Total 10 Marks)

a. A quadrilateral *ABCD* has vertices *A* (8, – 4), *B* (7, 1), *C* (– 5, 3) and *D* (− 4, − 6). If the area of triangle *ABC* is 29 units^2, then find the area of the quadrilateral *ABCD*. (5 Marks)

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.4.)

b. A line segment is formed by joining the points $P(-7, 1)$ and $Q(4, -3)$. Find the coordinates of a point Z that divides this line in the ratio 5 : 2

i. externally. (3 Marks)

ii. internally. (2 Marks)

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.4.)

c.

i. The lines l_1, l_2 and l_3 are the sides of a triangle. The lines l_1 and l_2 are given as $l_1 : y = 8x + 2$ and $l_2 : y = 3x + 3$. Find the angle of the triangle that is measured from l_1 to l_2. (3 Marks)

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ii. For the line $5x + y + 20 = 0$, determine if the points $(-4, -3)$ and $(5, 2)$ lie on the same side or opposite sides. (2 Marks)

Q.5. (Total 6 Marks)

A furniture company produces tables and chairs. The production process consists of carpentry and painting. Each table takes 5 hours in carpentry and 3 hours in painting department. Each chair requires 3 hours in carpentry and 1 hour in painting department. During a certain production period, 360 hours of carpentry time and 250 hours of painting time are available. Each table sold yields a profit of Rs 1,500 and each chair sold yields a profit of Rs 800.

Production Process	**Hours Required to make 1 Unit**		**Available Hours**
	Table	**Chair**	
Carpentry	5	3	360
Painting	3	1	250
Profit	1,500	800	

Suppose that *x* is the number of tables and *y* is the number of chairs produced in this production period. Formulate this problem as a linear programming problem by finding the

a. constraints. (5 Marks)

b. profit function. (1 Mark)

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(ATTEMPT EITHER PART a OR PART b OF Q.6.)

Q.6. (Total 6 Marks)

a. Find the equations of two tangents from the point (6, 5) to the circle $x^2 + y^2 = 25$.

(**Note:** The equation of tangent from any point to the circle $x^2 + y^2 = a^2$ is given by $y = mx + a\sqrt{1 + m^2}$, where *m* is the slope of the tangent.)

b. Find the equations of tangents to the circle $x^2 + y^2 + 2x - 4y + 4 = 0$ at the points on the circle whose *x*-coordinate is − 1.

(**Note:** The equation of tangent at a point (x_1, y_1) on the circle $x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c = 0$ is given by $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$.)

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.7.)

Q.7. (Total 10 Marks)

a. Find the equations of tangent and normal to the hyperbola $x^2 - 4y^2 + 10x + 24y + 5 = 0$ at (– 5, 1). (5 Marks)

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.7.)

b. For the hyperbola $x^2 - y^2 = 9$, find the

i. eccentricity. (1 Mark)

ii. coordinates of foci. (1 Mark)

iii. equations of directrices. (1 Mark)

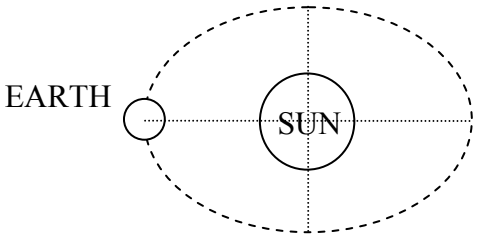
iv. lengths of latera recta. (1 Mark)

v. vertices. (1 Mark)

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.7.)

c. The earth is moving around the sun in an ELLIPTICAL path. During this motion, the shortest distance between their centres is 91 million miles while the farthest is 94.5 million miles.



If origin is considered as the centre of the sun, then the equation of the elliptical path will have the form $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$. Find the

i. values of a and b. (1 Mark)

ii. eccentricity of this ellipse. (2 Marks)

iii. lengths of latera recta. (1 Mark)

iv. distance between directrices of the earth orbit. (1 Mark)

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(ATTEMPT EITHER PART a OR PART b OF Q.8.)

Q.8. (Total 4 Marks)

a. Find the equation of the parabola $x^2 = 4y - 4$ with respect to new axes obtained by rotation of axes about the origin through an angle of $45°$.

b. A circle is defined as $(x-3)^2 + (y-2)^2 = 16$ with reference to *xy*-coordinate axes. Transform the equation of this circle with reference to (– 3, – 2) as new origin, such that the new axes remain parallel to the old axes. Also write the centre of circle with reference to the new origin (– 3, – 2).

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